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***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

ME/SE GE 15-1203 A ENGINEERING GRAPHICS

(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

(5 × 12 = 60)

- I. Construct a diagonal scale with RF 1/5000 showing meters and long enough to measure upto 1 kilometer. Mark a distance of 648 meters on it.

OR

- II. A circular disc 50 mm diameter rolls without slipping over a horizontal surface for half of its circumference. Then it climbs up a vertical wall and rolls another half of its circumference. Trace the path of point P which is initially on the vertical diameter, 32 mm away from the centre of the disc and below the initial point of contact of the disc on the horizontal surface.

- III. A tripod stand has three legs measuring 7 m, 8 m and 9 m all pivoted at a point O. The other ends of the legs are resting on level ground. The legs appear to be inclined at 120° to each other in the top view. The longest leg makes 75° to the ground. Draw the plan and elevation of the tripod and determine the inclination of the other two legs to the ground.

OR

- IV. A rhombus with diagonals 70 mm and 40 mm rests on the ground on a corner in such a way that the top view is a square with one side parallel to VP. Draw the plan and elevation of the rhombus in this position.

- V. A pentagonal prism 25 mm side of base and 70 mm long rests on the ground on an edge of base which is inclined at 30° to VP. The axis of the prism in this position makes 40° to HP. Draw plan and elevation of the prism in this position.

OR

- VI. A cylinder standing on the ground on its base is cut by a plane perpendicular to VP and inclined to HP such that the true shape is the biggest possible ellipse which has 90 mm major axis and 50 mm minor axis. Draw sectional plan, sectional elevation and true shape of section.

- VII. A lamp shade is made by cutting a cone of 60 mm diameter and 100 mm long by two cutting planes. One cutting plane is parallel to the base of the cone and passes through a point on the axis 70 mm above the base, while the other makes 35° to the base and passes through one of the extremities of base of the cone. Draw the development of the lamp shade.

OR

- VIII. A vertical square prism 50 mm side of base and 100 mm axis stands on HP on its base with an edge of base making 35° to VP. This is penetrated by another square prism 40 mm side of base and 100 mm long with its axis parallel to both planes and all lateral faces equally inclined to VP such that their axes intersect. Draw the three views showing the lines of intersection.

- IX. A ball 60 mm diameter rests inside an open box with inside dimensions 120 mm × 100 mm × 50 mm made of 5 mm thick plates. The ball touch one lateral surface of the box. Draw isometric projection of the box with the ball inside.

OR

- X. A hexagonal prism 25 mm side of base and 60 mm long lies in the ground on a rectangular face with axis perpendicular to picture plane. The nearest hexagonal face is 10 mm behind PP. The station point is 70 mm in front of PP, 30 mm above GP and the central plane passes through 50 mm to the right of the axis of the prism. Draw perspective view.

